
REPRESENTATION OF DISCRETE MAXIMUM ENTROPY DISTRIBUTIONS AS TENSOR NETWORKS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

July 13, 2026

ABSTRACT

This work investigates discrete distributions, which maximize the entropy with respect to linear constraints by the expectation of statistics. We introduce Computation-Activation Networks, which are tensor network formats representing generic maximum entropy distribution. The characterization exploits the face lattice of corresponding mean parameter polytopes to statistics, depending on which we provide rank bounds of the tensor network format. For boolean statistics we provide an interpretation based on propositional logics, which we utilize to construct statistics of generic lattice.

Contents

1	Introduction	2
1.1	Tensor Notation	2
1.2	The Maximum Entropy Problem	3
1.3	Computation-Activation Networks	3
1.4	Outlook	4
2	The mean polytope	4
2.1	Convex hull	4
2.2	Faces	4
2.3	Representation in the CP format	7
2.4	Examples of polytopes with elementary representable faces	7
2.5	Partition into Relative Interiors of Faces	8
3	Exponential Families	9
3.1	Exponential Families	9
3.2	Representation as elementary Computation-Activation Networks	9
3.3	Maximum entropy distributions	11
4	Generic maximum entropy distributions	11
4.1	Main result	12

4.2	Family of maximum entropy distributions	13
5	Boolean statistics	14
5.1	Characterization of elementary representable faces	14
5.2	Interpretation by propositional formulas	15
5.3	Construction of boolean statistics to polytopes	20
5.4	Further Examples of non-cubelike polytopes	20
6	Outlook	21

1 Introduction

The Shannon entropy of a distribution quantifies the amount of structure, or information, and appears therefore as an important concept across many scientific areas. By Shannons source code theorem Shannon (1948), entropy describes fundamental limits of information transfer. In statistical physics, founded on the ergodic assumptions, maximum entropy distributions such as Maxwell-Boltzmann distributions are the gold standard to model equilibrium distributions.

The principle of entropy maximization also appears naturally in learning problems. Extracting characteristics observed in data does not determine an estimated model yet. Choosing the distribution with the least structure among those reflecting the observed characteristics is a regularization principle. Here one can choose the Shannon entropy to quantify the negative amount of structure in the distribution and maximize the quantity with respect to the constraints. Furthermore, maximum entropy problems are the dual problems to convex maximum likelihood problems Koller and Friedman (2009).

Less obvious, studying generalizing maximum entropy distributions might be a key step towards a mathematically sound unification of logical and probabilistic models. Combining logical and probabilistic AI is a necessity in statistical relational AI, neuro-symbolic AI and explainable AI. Probability theory with exponential families typically focuses on distributions with maximal support. Logics on the contrary can be understood as a study of the support of models, marking the worlds consistent with a knowledge base. Logical and probabilistic models can be treated in the tensor network formalism, which thus serves as a unifying representation language Goessmann et al. (2026).

Maximum entropy principles appear further in reinforcement learning Yoon et al. (2024); Song et al. (2025), AGI Miotto and Monacelli (2025) and Bayesian NN Rathnakumar et al. (2026).

1.1 Tensor Notation

We introduce tensors based on index variables X_k to each of the $k \in [d] = \{0, \dots, d-1\}$ axes, each taking values in $[m_k] = \{0, \dots, m_k - 1\}$, where $m_k \in \mathbb{N}$ is the dimension of the variable. Tensors are then maps

$$\tau [X_{[d]}] : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}$$

from indices $x_{[d]} = (x_0, \dots, x_{d-1}) \in \bigtimes_{k \in [d]} [m_k]$ to their coordinates $\tau [X_{[d]} = x_{[d]}] \in \mathbb{R}$. One-hot encodings of index tuples $x_{[d]}$ are specific tensors $\epsilon_{x_{[d]}} [X_{[d]}]$ such that $\epsilon_{x_{[d]}} [X_{[d]} = \tilde{x}_{[d]}] = 1$ if $\tilde{x}_{[d]} = x_{[d]}$ and 0 else.

Given a hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ of an arbitrary node set $\mathcal{V}$ and hyperedges $e \in \mathcal{E}$ being subsets of $\mathcal{V}$, we define decorating tensor network as follows. Each node $v \in \mathcal{V}$ gets decorated by a variable X_v of dimension m_v and each hyperedge $e \in \mathcal{E}$ by a tensor $\tau^e [X_e]$ with the index variables to the contained nodes. The set of decorating tensors then builds the tensor network $\tau^{\mathcal{G}}$ on the hypergraph. A contraction of the tensor network keeping the nodes $\mathcal{U} \subset \mathcal{V}$ open is then the tensor $\langle \tau^{\mathcal{G}} \rangle_{[X_{\mathcal{U}}]}$ with variables $X_{\mathcal{U}}$ and coordinates to the indices $x_{\mathcal{U}}$ by

$$\langle \tau^{\mathcal{G}} \rangle_{[X_{\mathcal{U}} = x_{\mathcal{U}}]} = \sum_{x_{\mathcal{V} \setminus \mathcal{U}}} \prod_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

1.2 The Maximum Entropy Problem

Given a non-negative and non-vanishing base measure $\nu [X_{[d]}]$, a probability distribution is a non-negative tensor $\mathbb{P} [X_{[d]}]$ such that

$$\langle \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]} = 1.$$

We denote the set of such distributions by

$$\Lambda^\nu = \left\{ \mathbb{P} [X_{[d]}] : \mathbb{P} [X_{[d]}] \geq 0 [X_{[d]}], \langle \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]} = 1, \right. \\ \left. \forall x_{[d]} \in \times_{k \in [d]} [m_k] \nu [X_{[d]} = x_{[d]}] = 0 \Rightarrow \mathbb{P} [X_{[d]} = x_{[d]}] = 0 \right\}.$$

Note that we here restrict the support of the tensor $\mathbb{P} [X_{[d]}]$ to be included in the support of the base measure, to avoid ambiguity in the representation.

The entropy of a distribution relative to the base measure ν is

$$\mathbb{H}_\nu [\mathbb{P} [X_{[d]}]] = - \langle \mathbb{P} [X_{[d]}], \ln [\mathbb{P} [X_{[d]}]] , \nu [X_{[d]}] \rangle_{[\emptyset]}.$$

Note that this quantity depends on the chosen base measure.

The mean parameter of a distribution $\mathbb{P} [X_{[d]}]$ to a statistic $t : \times_{k \in [d]} [m_k] \rightarrow \times_{s \in [n]} [p_s]$ is the vector $\mu [L] \in \mathbb{R}^p$ with the coordinates

$$\mu [L = \ell] = \mathbb{E} [f_\ell] = \langle f_\ell [X_{[d]}], \mathbb{P} [X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]}.$$

We express the computation of the mean parameter in the contraction of the selection encoding $\sigma^t [X_{[d]}, L]$ of t

$$\mu [L] = \langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]}.$$

The maximum entropy problem given a mean parameter $\mu^* [L]$ is stated by

$$\operatorname{argmax}_{\mathbb{P} [X_{[d]}] \in \Lambda^\nu} \mathbb{H}_\nu [\mathbb{P} [X_{[d]}]] \quad \text{subject to} \quad \langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L], \nu [X_{[d]}] \rangle_{[L]} = \mu^* [L] \quad (\mathbf{P}_{t, \mu, \nu})$$

1.3 Computation-Activation Networks

We now introduce with Computation-Activation Networks families of tensor networks representing, as we will show in this work, the solution of any instance of Problem $\mathbf{P}_{t, \mu, \nu}$. Given a statistic $t : \times_{k \in [d]} [m_k] \rightarrow \times_{s \in [n]} [p_s]$ we define its basis encoding tensor

$$\beta^t [Y_{[p]}, X_{[d]}] = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \epsilon_{t(x_{[d]})} [Y_{[p]}] \otimes \epsilon_{x_{[d]}} [X_{[d]}].$$

A computation network is any representation of $\beta^t [Y_{[p]}, X_{[d]}]$ as a tensor network. These can be constructed in the case statistics being a composition of connective functions.

An activation tensor is $\tau [Y_{[p]}]$ and the corresponding Computation-Activation Network with respect to the statistic t and the base measure $\nu [X_{[d]}]$ the tensor

$$\mathbb{P} [X_{[d]}] = \frac{\langle \beta^t [Y_{[p]}, X_{[d]}], \tau [Y_{[p]}] \rangle_{[X_{[d]}]}}{\langle \beta^t [Y_{[p]}, X_{[d]}], \tau [Y_{[p]}], \nu [X_{[d]}] \rangle_{[\emptyset]}}.$$

We are interested in decomposition formats of $\tau [Y_{[p]}]$, where we use sets of tensor networks $\mathcal{T}^\mathcal{G}$ on a hypergraph $\mathcal{G}$.

Definition 1. *The family of by t and a $\mathcal{G}$ computable distributions are*

$$\Lambda^{t, \mathcal{G}, \nu} = \left\{ \frac{\langle \tau [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}] \rangle_{[X_{[d]}]}}{\langle \tau [Y_{[p]}], \beta^t [Y_{[p]}, X_{[d]}], \nu [X_{[d]}] \rangle_{[\emptyset]}} : \tau [Y_{[p]}] \in \mathcal{T}^\mathcal{G} \right\}.$$

1.4 Outlook

To prepare for the presentation of our main results we introduce in section Sect. 2 the mean polytope, its faces, and their tensor network representations. In Sect. 3 we study exponential families as classes of elementary Computation-Activation Networks and show that they are the maximum entropy distributions to mean parameters in the relative interior of the mean polytope. This result is generalized in Sect. 4 where we characterize generic maximum entropy distributions based on Computation-Activation Networks. In Sect. 5 we investigate consequences of these results in case of boolean statistics and connect them to propositional logics.

2 The mean polytope

The mean polytope is the set of possible mean parameters given a by base measure representable distribution in Λ^ν . We define it as

$$\mathcal{M}_{t,\nu} := \left\{ \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} : \mathbb{P}[X_{[d]}] \in \Lambda^\nu \right\},$$

where we denote by Λ^ν the set of all probability distributions representable with respect to ν , and by $\sigma^t[X_{[d]}, L]$ the selection encoding of the statistic (see introduction).

2.1 Convex hull

First of all, we characterize the mean polytope as the convex hull of the slices to the selection encoding.

Lemma 1. *For any statistic t and base measure ν , the set of mean parameters is the convex hull of the set*

$$\mathcal{N}_{t,\nu} := \{ \sigma^t[X_{[d]} = x_{[d]}, L] : x_{[d]} \in \bigtimes_{k \in [d]} [m_k], \nu[X_{[d]} = x_{[d]}] \neq 0 \}$$

that is $\mathcal{M}_{t,\nu} = \text{conv}(\mathcal{N}_{t,\nu})$.

Proof. This follows from

$$\Lambda^\nu = \text{conv} \left(\frac{1}{\nu[X_{[d]} = x_{[d]}]} \cdot \epsilon_{x_{[d]}}[X_{[d]}] : x_{[d]} \in \bigtimes_{k \in [d]} [m_k], \nu[X_{[d]} = x_{[d]}] \neq 0 \right).$$

and for any vertex of this simplex we have

$$\left\langle \frac{1}{\nu[X_{[d]} = x_{[d]}]} \cdot \epsilon_{x_{[d]}}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \right\rangle_{[L]} = \sigma^t[X_{[d]} = x_{[d]}, L].$$

□

2.2 Faces

Let us now continue with the investigation of the faces of the mean parameter polytope.

Definition 2. *We say that the convex hull of a subset $\mathcal{N}_{t,\nu}^\mathcal{F} \subset \mathcal{N}_{t,\nu}$ is a face of a mean polytope $\mathcal{M}_{t,\nu}$, if and only if there is a face normal vector $\theta_\mathcal{F}[L] \in \mathbb{R}^p$ such that*

$$\mathcal{N}_{t,\nu}^\mathcal{F} = \text{argmax}_{\mu[L] \in \mathcal{N}_{t,\nu}} \langle \mu[L], \theta_\mathcal{F}[L] \rangle_{[\emptyset]}.$$

We denote the face as $\mathcal{F} = \text{conv}(\mathcal{N}_{t,\nu}^\mathcal{F})$. The set of all faces of the mean polytope $\mathcal{M}_{t,\nu}$ is denoted by $L(\mathcal{M}_{t,\nu})$.

The set $L(\mathcal{M}_{t,\nu})$ is partially ordered by the inclusion operation and is called a lattice Ziegler (2013).

We notice, that each face itself is a convex polytope. What is more, we can characterize these as mean parameter polytopes with respect to refined base measures to be defined next.

Definition 3. *The to a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ refined base measure is the tensor $\nu^{t,\mathcal{F}}[X_{[d]}]$ with coordinates to $x_{[d]} \in \bigtimes_{k \in [d]} [m_k]$ by*

$$\nu^{t,\mathcal{F}}[X_{[d]}] = \begin{cases} \nu[X_{[d]} = x_{[d]}] & \text{if } t(x_{[d]}) \in \mathcal{N}_{t,\nu}^\mathcal{F} \\ 0 & \text{else} \end{cases}.$$

We now specify the mean parameter polytope to any face using the face measure as a refinement of the base measure.

Lemma 2. *For any face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have*

$$\mathcal{F} = \mathcal{M}_{t,\nu^t,\mathcal{F}}.$$

Proof. For any $x_{[d]} \in \times_{k \in [d]} [m_k]$ we have $\nu^{t,\mathcal{F}}[X_{[d]} = x_{[d]}] \neq 0$ if and only if $t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and $\nu[X_{[d]} = x_{[d]}] \neq 0$. Thus we have

$$\begin{aligned} \mathcal{F} &= \text{conv}(\mathcal{N}_{t,\nu}^{\mathcal{F}}) \\ &= \text{conv}(\sigma^t[X_{[d]} = x_{[d]}, L] : t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}, \nu[X_{[d]} = x_{[d]}] \neq 0) \\ &= \text{conv}(\sigma^t[X_{[d]} = x_{[d]}, L] : \nu^{t,\mathcal{F}}[X_{[d]} = x_{[d]}] \neq 0) \\ &= \mathcal{M}_{t,\nu^t,\mathcal{F}}. \end{aligned} \quad \square$$

Representability of a distribution with respect to face measures is an equivalent condition for the mean parameter of a distribution to be on a face, as we show next.

Lemma 3. *If and only if for a distribution $\mathbb{P}[X_{[d]}] \in \Lambda^\nu$ and a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have*

$$\langle \sigma^t[X_{[d]}, L], \mathbb{P}[X_{[d]}], \nu[X_{[d]}] \rangle_{[L]} \in \mathcal{F},$$

then

$$\mathbb{P}[X_{[d]}] \in \Lambda^{\nu^{t,\mathcal{F}}}.$$

Proof. For a distribution $\mathbb{P}[X_{[d]}] \in \Lambda^\nu$ to be in $\Lambda^{\nu^{t,\mathcal{F}}}$, it is enough to show that $\mathbb{P}[X_{[d]}]$ is only supported at $\nu^{t,\mathcal{F}}[X_{[d]}]$. We have

$$\mu[L] = \sum_{x_{[d]}} \mathbb{P}[X_{[d]} = x_{[d]}] \cdot \nu[X_{[d]} = x_{[d]}] \cdot \gamma^t[X_{[d]} = x_{[d]}, L].$$

Let now $\theta_{\mathcal{F}}[L]$ be a face normal to the face $\mathcal{F}$. We then have

$$\langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]} = \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \mathbb{P}[X_{[d]} = x_{[d]}] \cdot \nu[X_{[d]} = x_{[d]}] \cdot \langle \gamma^t[X_{[d]} = x_{[d]}, L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]}.$$

Now if and only if $\mathbb{P}[X_{[d]} = x_{[d]}] \cdot \nu[X_{[d]} = x_{[d]}]$ is supported only for $x_{[d]}$ with $t(x_{[d]}) \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ we have that

$$\langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]} = \max_{\mu[L] \in \mathcal{M}_{t,\nu}} \langle \mu[L], \theta_{\mathcal{F}}[L] \rangle_{[\emptyset]}$$

which is equal to $\mu[L] \in \mathcal{F}$. Thus, if and only if $\mu[L] \in \mathcal{F}$ then $\mathbb{P}[X_{[d]}]$ is only supported at $x_{[d]}$ in the support of $\nu^{t,\mathcal{F}}[X_{[d]}]$. $\square$

Let us now investigate tensor network representations of refined base measures, based on the basis encoding β^t of a statistic.

Theorem 1. *For any face $\mathcal{F}$ of $\mathcal{M}$ we have*

$$\nu^{t,\mathcal{F}}[X_{[d]}] = \langle \beta^t[Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}}[Y_{[p]}], \nu[X_{[d]}] \rangle_{[X_{[d]}]}$$

where

$$\kappa^{\mathcal{F}}[Y_{[p]}] = \sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_{v[L]}[Y_{[p]}].$$

Proof. For any $v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}$ the tensor

$$\tau^v[X_{[d]}] = \langle \beta^t[Y_{[p]}, X_{[d]}], \epsilon_{v[L]}[Y_{[p]}] \rangle_{[X_{[d]}]}$$

is the indicator of the preimage of μ under σ^t . Since preimages of the elements in $\mathcal{N}_{t,\nu}^{\mathcal{F}}$ are disjoint, the support of $\tau^v [X_{[d]}]$ is disjoint and their sum

$$\sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \tau^v [X_{[d]}]$$

is the indicator of the preimage of $\mathcal{F}$ under σ^t . The face measure obeys thus

$$\begin{aligned} \nu^{t,\mathcal{F}} [X_{[d]}] &= \left\langle \left(\sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \tau^v [X_{[d]}] \right), \nu [X_{[d]}] \right\rangle_{[X_{[d]}]} \\ &= \sum_{v[L] \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \langle \beta^t [Y_{[p]}], X_{[d]} \rangle, \epsilon_{v[L]} [Y_{[p]}], \nu [X_{[d]}] \rangle_{[X_{[d]}]} \\ &= \langle \beta^t [Y_{[p]}], X_{[d]} \rangle, \kappa^{\mathcal{F}} [Y_{[p]}], \nu [X_{[d]}] \rangle_{[X_{[d]}]} \quad \square \end{aligned}$$

We now investigate the representation of face measures by Computation-Activation Networks.

Definition 4. Let $\mathcal{G}$ be a hypergraph which nodes include $[p]$. We say that a face $\mathcal{F}$ is representable by $\mathcal{G}$ if and only if there is a set $\mathcal{U}$ of basis vectors with $\mathcal{U} \cap \mathcal{N}_{t,\nu} = \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and there is a tensor network $\kappa^{\mathcal{G}}$ with respect to the hypergraph $\mathcal{G}$ such that

$$\langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]} = \sum_{v \in \mathcal{U}} \epsilon_v [Y_{[p]}] .$$

We call any such tensor network a face activating tensor network.

Lemma 4. If any only if a face is representable by a hypergraph $\mathcal{G}$ we have

$$\frac{1}{\langle \nu^{t,\mathcal{F}} [X_{[d]}] \rangle_{[\emptyset]}} \cdot \nu^{t,\mathcal{F}} [X_{[d]}] \in \Lambda^{t,\mathcal{G},\nu} .$$

Proof. For any $\mathcal{U}$ with $\mathcal{U} : \mathcal{U} \cap \mathcal{N}_{t,\nu} = \mathcal{N}_{t,\nu}^{\mathcal{F}}$ and tensor network $\kappa^{\mathcal{G}}$ respecting the assumptions of Def. 4 we have

$$\begin{aligned} \left\langle \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]}, \beta^t [Y_{[p]}], X_{[d]} \right\rangle_{[X_{[d]}]} &= \left\langle \sum_{v \in \mathcal{U}} \epsilon_v [Y_{[p]}], \beta^t [Y_{[p]}], X_{[d]} \right\rangle_{[X_{[d]}]} \\ &= \left\langle \sum_{v \in \mathcal{N}_{t,\nu}^{\mathcal{F}}} \epsilon_v [Y_{[p]}], \beta^t [Y_{[p]}], X_{[d]} \right\rangle_{[X_{[d]}]} . \end{aligned}$$

Here we used in the second equation that only the vertices in $\mathcal{N}_{t,\nu}$ are in the image of t . It follows, that

$$\frac{1}{\langle \nu^{t,\mathcal{F}} [X_{[d]}] \rangle_{[\emptyset]}} \cdot \nu^{t,\mathcal{F}} [X_{[d]}] = \frac{\left\langle \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]}, \beta^t [Y_{[p]}], X_{[d]} \right\rangle_{[X_{[d]}]}}{\left\langle \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]}, \beta^t [Y_{[p]}], X_{[d]} \rangle, \nu [X_{[d]}] \right\rangle_{[\emptyset]}}$$

and thus $\nu^{t,\mathcal{F}} [X_{[d]}] \in \Lambda^{t,\mathcal{G},\nu}$. $\square$

Let us now investigate, which normalized face measures can be computed using t and a hypergraph $\mathcal{G}$.

Example 1 (Vertices). Vertices $\mathcal{F} = \{v [L]\}$ are proper faces of affine dimension 0, that is they consist in single vectors $v [L]$. Since all vertices are in the image $\mathcal{N}_{t,\nu}$, there exists an index tuple $x_{[d]} \in \mathcal{X}$ such that $\nu [X_{[d]} = x_{[d]}] = 1$ and

$$v [L] = \sigma^t [X_{[d]} = x_{[d]}, L] .$$

Then $\kappa^{\mathcal{F}} [Y_{[p]}]$ is the one-hot encoding of the vector $\sigma^t [X_{[d]} = x_{[d]}, L]$, that is

$$\kappa^{\mathcal{F}} [Y_{[p]}] = \epsilon_{\sigma^t [X_{[d]} = x_{[d]}, L]} [Y_{[p]}] .$$

Such tensors are examples of elementary tensors corresponding to the elementary hypergraph

$$\text{EL} = ([p], \{\{\ell\} : \ell \in [p]\}) .$$

In particular, the activation core is elementary and the corresponding refined base measures are in $\Lambda^{t,\text{EL},\nu}$.

While vertices are the minimal non-vanishing faces in the face-lattice (see Ziegler (2013)), we now show that also the maximal face, namely the polytope itself, is representable with respect to the elementary hypergraph EL.

Example 2 (Maximal face). *The maximal face $\mathcal{F}^\emptyset = \mathcal{M}_{t,\nu}$ coincides with the mean polytope itself and is given by the choice $\theta_\emptyset [L] = 0 [L]$. In this case the corresponding activation tensor to the face measure is trivial, that is*

$$\kappa^\emptyset [Y_{[p]}] = \mathbb{I} [Y_{[p]}] = \bigotimes_{\ell \in [p]} \mathbb{I} [Y_\ell] .$$

$\kappa^\emptyset$ is elementary and the maximal face is always representable by an elementary activation.

2.3 Representation in the CP format

Extending Example 1, we can provide a coarse estimation of the hypergraph $\mathcal{G}$ required to decompose $\kappa^\mathcal{F}$ for generic faces $\mathcal{F}$. To this end, we use the CP format describing tensor networks on the CP hypergraph

$$\text{CP} = ([p] \cup \{I\}, \{\{\ell, I\} : \ell \in [p]\}) .$$

The dimension of the variable I is called the hidden rank of the format.

Lemma 5. *Any face $\mathcal{F}$ is representable by a CP graph with hidden rank*

$$r \leq \min (|\mathcal{N}_{t,\nu}^\mathcal{F}|, |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^\mathcal{F}| + 1) .$$

Proof. We show the claim by constructing two face activating tensor networks to $\mathcal{F}$ in a CP hypergraph with hidden rank $|\mathcal{N}_{t,\nu}^\mathcal{F}|$ and in a CP hypergraph with hidden rank $|\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^\mathcal{F}| + 1$. To show the first representation we enumerate the vertices by a variable I with dimension $r = |\mathcal{N}_{t,\nu}^\mathcal{F}|$, i.e. $\mathcal{N}_{t,\nu}^\mathcal{F} = \{v^i [L] : i \in [n]\}$. Then we define for $\ell \in [p]$ core tensors $\tau^\ell [Y_\ell, I]$

$$\tau^\ell [Y_\ell, I = i] = \epsilon_{v^i [L=\ell]} [Y_\ell] .$$

Then we have

$$\langle \{\tau^\ell [Y_\ell, I] : \ell \in [p]\} \rangle_{[Y_{[p]}]} = \sum_{v \in \mathcal{N}_{t,\nu}^\mathcal{F}} \epsilon_v [Y_{[p]}]$$

and thus have found an activation tensor network in a CP graph with hidden rank $|\mathcal{N}_{t,\nu}^\mathcal{F}|$ representing the face $\mathcal{F}$.

We continue with the second representation, for which we enumerate the set $\mathcal{N}_{t,\nu}/\mathcal{N}_{t,\nu}^\mathcal{F}$ by $v^i [L]$ where $i \in [|\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^\mathcal{F}|]$. We define variable I with dimension $r = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^\mathcal{F}| + 1$ and define for $\ell \in [p]$ core tensors

$$\tau^\ell [Y_\ell, I = i] = \begin{cases} -\epsilon_{v^i [L=\ell]} [Y_\ell] & \text{if } i < |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^\mathcal{F}| \\ \mathbb{I} [Y_\ell] & \text{if } i = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^\mathcal{F}| \end{cases} .$$

We then have

$$\begin{aligned} \langle \{\tau^\ell [Y_\ell, I] : \ell \in [p]\} \rangle_{[Y_{[p]}]} &= \mathbb{I} [Y_{[p]}] - \sum_{v \in \mathcal{N}_{t,\nu}/\mathcal{N}_{t,\nu}^\mathcal{F}} \epsilon_v [Y_{[p]}] \\ &= \sum_{v \in \mathcal{N}_{t,\nu}^\mathcal{F}} \epsilon_v [Y_{[p]}] + \sum_{v \in (\mathcal{N}_{t,\nu}/\mathcal{N}_{t,\nu}^\mathcal{F})/\mathcal{N}_{t,\nu}} \epsilon_v [Y_{[p]}] . \end{aligned}$$

We have thus found a face activating tensor network for $\mathcal{F}$ in a CP format with hidden rank $r = |\mathcal{N}_{t,\nu}| - |\mathcal{N}_{t,\nu}^\mathcal{F}| + 1$. $\square$

2.4 Examples of polytopes with elementary representable faces

We show in the following example, that all faces of a hypercube and a simplex are representable by elementary activation tensors.

Example 3 (Hypercube). *In cases where $X_{[d]}$ are boolean and we have $\ell \in [p]$ features*

$$f_\ell [X_{[d]} = x_{[d]}] = x_\ell$$

the mean polytope is the hypercube

$$\mathcal{M}_{\{f_\ell : \ell \in [p]\}, \mathbb{I}} = [0, 1]^p.$$

The non-empty faces in the face lattice $L([0, 1]^p)$ can be enumerated by subsets $\mathcal{U} \subset [p]$ and indices $y_{\mathcal{U}} \in \times_{\ell \in \mathcal{U}} [2]$ and represented by the cartesian products

$$\mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})} = \bigtimes_{\ell \in [p]} \mathcal{I}^{\ell, (\mathcal{U}, y_{\mathcal{U}})}$$

where

$$\mathcal{I}^{\ell, (\mathcal{U}, y_{\mathcal{U}})} = \begin{cases} [0, 1] & \text{if } \ell \notin \mathcal{U} \\ \{y_\ell\} & \text{if } \ell \in \mathcal{U} \end{cases}.$$

Each of these faces can be represented with respect to the elementary graph EL, namely by the tensor product of leg vectors

$$\kappa^{\ell, (\mathcal{U}, y_{\mathcal{U}})} [Y_\ell] = \begin{cases} \mathbb{I} [Y_\ell] & \text{if } \ell \notin \mathcal{U} \\ \epsilon_{y_\ell} [Y_\ell] & \text{if } \ell \in \mathcal{U} \end{cases}.$$

Example 4 (Simplices). The $(p-1)$ -dimensional standard simplex $\mathcal{M}_{\Delta, p-1}$ in $[0, 1]^p$ is the convex hull of the vertex sets

$$\mathcal{N} = \{\epsilon_\ell [L] : \ell \in [p]\}.$$

The simplex is the mean polytope of the p -dimensional statistic with

$$t_\ell : [p] \rightarrow [2] \quad , \quad t_\ell [\tilde{\ell}] = \begin{cases} 1 & \text{if } \tilde{\ell} = \ell \\ 0 & \text{if } \tilde{\ell} \neq \ell \end{cases}.$$

The simplex represents all distributions with respect to the trivial base measure $\mathbb{I}$ which coordinates are all 1. When flattened to a vector, any tensor in $\Lambda^{\mathbb{I}}$ is an element of $\mathcal{M}_{\Delta, p-1}$ where $p = \prod_{k \in [d]} m_k$.

The face lattice of the simplex consists is enumerated by subsets of $[p]$ as

$$L(\mathcal{M}_{\Delta, p-1}) = \{\mathcal{F}^{\mathcal{U}} : \mathcal{U} \subset [p]\}$$

where the faces are

$$\mathcal{F}^{\mathcal{U}} = \text{conv}(\epsilon_\ell [L] : \ell \in \mathcal{U})$$

and therefore itself $(|\mathcal{U}|-1)$ -dimensional standard simplices. The partial order of the faces coincides with the inclusion order of subsets $\mathcal{U}$.

2.5 Partition into Relative Interiors of Faces

Let us now introduce relative interiors, which enables us to find disjoint partitions of the mean polytope.

Definition 5 (Relative Interior). The relative interior $(\mathcal{U})^\circ$ of an arbitrary set $\mathcal{U} \subset \mathbb{R}^p$ is the interior of $\mathcal{U}$ in the affine subspace $\text{conv}(\mathcal{U})$.

Lemma 6. Any polytope is a disjoint union of the relative interiors of its faces, that is

$$\mathcal{M}_{t, \nu} = \bigcup_{\mathcal{F} \in L(\mathcal{M}_{t, \nu})} (\mathcal{F})^\circ.$$

Proof. For any $\mu \in \mathcal{M}_{t, \nu}$ we find a face such that $\mu \in \mathcal{F}$. If $\mu \notin (\mathcal{F})^\circ$, then there is a face $\tilde{\mathcal{F}} \subset \mathcal{F}$ of smaller affine dimension such that $\mu \in \tilde{\mathcal{F}}$. We then redo the above argument for $\mathcal{F} = \tilde{\mathcal{F}}$ until we arrive at a face with $\mu \in (\mathcal{F})^\circ$. We always reach such a face, since the affine dimension of the faces is decreasing and lower bounded by 0, and the faces of affine dimension 0 are vertices for which $\mathcal{F} = (\mathcal{F})^\circ$. $\square$

Definition 6. To each $\mu \in \mathcal{M}_{t, \nu}$ we denote the unique face $\mathcal{F}$ with $\mu \in (\mathcal{F})^\circ$ by $\mathcal{F}(\mu)$.

3 Exponential Families

It is well known that exponential families are specific maximum entropy distributions. However, not any maximum entropy distribution is an exponential family member. To prepare for the characterization of generic maximum entropy distributions, we will in this section represent exponential family members as Computation-Activation Networks. We further show that a maximum entropy distribution is an exponential family member, if and only if its mean parameter is in the relative interior of the mean polytope.

3.1 Exponential Families

Definition 7 (Exponential Family). *Given a statistic function*

$$t : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}^p$$

and a base measure

$$\nu : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}_0^+$$

with $\langle \nu \rangle_{[\emptyset]} \neq 0$, the set $\Gamma^{t, \nu} = \{\mathbb{P}^{(t, \theta, \nu)} : \theta[L] \in \mathbb{R}^p\} \subset \Lambda^\nu$ of probability distributions

$$\mathbb{P}^{(t, \theta, \nu)} [X_{[d]}] = \frac{\exp \left[\langle \sigma^t [X_{[d]}, L], \theta [L] \rangle_{[X_{[d]}]} \right]}{\left\langle \exp \left[\langle \sigma^t [X_{[d]}, L], \theta [L] \rangle_{[X_{[d]}]} \right], \nu [X_{[d]}] \right\rangle_{[\emptyset]}}$$

is called the exponential family to t .

To present a tensor network representation, we introduce image interpretation maps

$$I_\ell : [|\text{im}(t_\ell)|] \rightarrow \text{im}(t_\ell),$$

which enumerate the possible values of each feature. We treat these maps as tensors with in a variable Y_ℓ with values in $[|\text{im}(t_\ell)|]$.

3.2 Representation as elementary Computation-Activation Networks

Theorem 2 (Exponential Families are elementary Computation-Activation Networks). *Given any base measure ν and a sufficient statistic t we enumerate for each coordinate $\ell \in [p]$ the image $\text{im}(t_\ell)$ by a variable Y_ℓ taking values in $[|\text{im}(t_\ell)|]$, given an interpretation map*

$$I_\ell : [|\text{im}(t_\ell)|] \rightarrow \text{im}(t_\ell).$$

For any canonical parameter vector $\theta[L] \in \mathbb{R}^p$ we build the activation cores $\alpha^{\ell, \theta} [Y_\ell]$ for each coordinate $y_\ell \in [|\text{im}(t_\ell)|]$ by

$$\alpha^{\ell, \theta} [Y_\ell = y_\ell] = \exp [\theta [L = \ell] \cdot I_\ell(y_\ell)]$$

and have (see Figure 1)

$$\mathbb{P}^{(t, \theta, \nu)} [X_{[d]}] = \frac{\langle \{\beta^t [Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta} [Y_\ell] : \ell \in [p]\} \rangle_{[X_{[d]}]}}{\langle \{\beta^t [Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta} [Y_\ell] : \ell \in [p]\} \cup \{\nu [X_{[d]}]\} \rangle_{[\emptyset]}}.$$

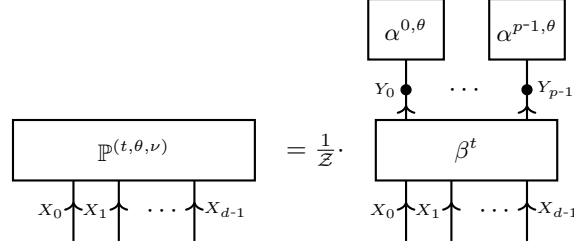


Figure 1: Representation of a member in an exponential family by a Computation-Activation Network with elementary activation. Here $\mathcal{Z} \in \mathbb{R}$ is a normalization constant.

Proof. For each $x_{[d]} \in \times_{k \in [d]} [m_k]$ we have

$$\begin{aligned}
 & \langle \{\beta^t [Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta} [Y_\ell] : \ell \in [p]\} \rangle_{[X_{[d]}=x_{[d]}]} \\
 &= \prod_{\ell \in [p]} \langle \beta^{t_\ell} [Y_\ell, X_{[d]} = x_{[d]}], \alpha^{\ell, \theta} [Y_\ell] \rangle_{[\emptyset]} \\
 &= \prod_{\ell \in [p]} \exp [\theta [L = \ell] \cdot I_\ell(t_\ell [x_{[d]}])] \\
 &= \exp \left[\sum_{\ell \in [p]} \theta [L = \ell] \cdot I_\ell(t_\ell [x_{[d]}]) \right] \\
 &= \exp \left[\langle \sigma^t [X_{[d]}, L], \theta [L] \rangle_{[X_{[d]}=x_{[d]}]} \right].
 \end{aligned}$$

Therefore we have

$$\langle \{\beta^t [Y_{[p]}, X_{[d]}]\} \cup \{\alpha^{\ell, \theta} [Y_\ell] : \ell \in [p]\} \rangle_{[X_{[d]}]} = \exp \left[\langle \sigma^t [X_{[d]}, L], \theta [L] \rangle_{[X_{[d]}]} \right].$$

The claim follows, since this implies that also the contraction of both sides with $\nu [X_{[d]}]$ is equivalent. $\square$

We will use the following well known property (see e.g. Brown (1987)), that there is a one-to-one map between the canonical parameters and the mean parameters.

Lemma 7. *The set of mean parameters of the members of an exponential family is the relative interior of the mean polytope. For $\theta, \tilde{\theta} \in \mathbb{R}^p$ with*

$$\langle \mathbb{P}^{t, \theta, \nu} [X_{[d]}], \sigma^t [X_{[d]}, L], \nu [X_{[d]}] \rangle_{[L]} = \langle \mathbb{P}^{t, \tilde{\theta}, \nu} [X_{[d]}], \sigma^t [X_{[d]}, L], \nu [X_{[d]}] \rangle_{[L]}$$

we furthermore have $\mathbb{P}^{t, \theta, \nu} [X_{[d]}] = \mathbb{P}^{t, \tilde{\theta}, \nu} [X_{[d]}]$.

Proof. See The 3.3 in Wainwright and Jordan (2008). $\square$

Based on this property we define the forward and backward mappings to an exponential family.

Definition 8. *The forward map of an exponential family is the map*

$$F^{(t, \nu)} : \mathbb{R}^p \rightarrow (\mathcal{M}_{t, \nu})^\circ$$

defined as

$$F^{(t, \nu)}(\theta) = \langle \mathbb{P}^{t, \theta, \nu} [X_{[d]}], \sigma^t [X_{[d]}, L], \nu [X_{[d]}] \rangle_{[L]}$$

Any map $B^{(t, \nu)} : (\mathcal{M}_{t, \nu})^\circ \rightarrow \mathbb{R}^p$ with $F^{(t, \nu)}(B^{(t, \nu)}(\mu)) = \mu$ for all $\mu \in (\mathcal{M}_{t, \nu})^\circ$ is called a backward map.

3.3 Maximum entropy distributions

We now show that exponential families are maximum entropy distributions.

Lemma 8. *Any member of an exponential family $\Gamma^{t,\nu}$ is a maximum entropy distribution with respect to the moment constraint on t and the base measure ν .*

Proof. Let $\mathbb{P}^{(t,\theta,\nu)}$ be an exponential family member with canonical parameter θ and let $\mu[L]$ be the corresponding mean parameter. Let $\tilde{\mathbb{P}}[X_{[d]}]$ be any feasible distribution for the maximum entropy problem with respect to the moment constraint on t by $\mu[L]$ and the base measure ν . We also have $\langle \tilde{\mathbb{P}}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} = \mu[L]$ and thus

$$\begin{aligned} \mathbb{H}_\nu[\tilde{\mathbb{P}}, \mathbb{P}^{(t,\theta,\nu)}] &= -\left\langle \tilde{\mathbb{P}}[X_{[d]}], \ln[\mathbb{P}^{(t,\theta,\nu)}[X_{[d]}]], \nu[X_{[d]}] \right\rangle_{[\emptyset]} \\ &= -\left\langle \tilde{\mathbb{P}}[X_{[d]}], \sigma^t[X_{[d]}, L], \theta[L], \nu[X_{[d]}] \right\rangle_{[\emptyset]} + A^{(t,\nu)}(\theta) \\ &= -\langle \theta[L], \mu[L] \rangle_{[\emptyset]} + A^{(t,\nu)}(\theta) \\ &= \mathbb{H}_\nu[\mathbb{P}^{(t,\theta,\nu)}]. \end{aligned}$$

With the Gibbs inequality we have if $\tilde{\mathbb{P}} \neq \mathbb{P}^{(t,\theta,\nu)}$

$$\mathbb{H}_\nu[\mathbb{P}^{(t,\theta,\nu)}] - \mathbb{H}_\nu[\tilde{\mathbb{P}}] = \mathbb{H}_\nu[\tilde{\mathbb{P}}, \mathbb{P}^{(t,\theta,\nu)}] - \mathbb{H}_\nu[\tilde{\mathbb{P}}] > 0$$

and thus $\mathbb{H}_\nu[\tilde{\mathbb{P}}] < \mathbb{H}_\nu[\mathbb{P}^{(t,\theta,\nu)}]$. Therefore, any feasible $\tilde{\mathbb{P}}$ different from $\mathbb{P}^{(t,\theta,\nu)}$ have less entropy than $\mathbb{P}^{(t,\theta,\nu)}$ and $\mathbb{P}^{(t,\theta,\nu)}$ is a maximum entropy distribution. $\square$

Together with Lem. 7 stating the existence of an exponential family member for any mean parameter in the relative interior of the mean polytope we arrive at the following theorem.

Theorem 3. *If and only if $\mu[L]$ is in the relative interior of $\mathcal{M}_{t,\nu}$, then the unique solution of the maximum entropy problem is the distribution*

$$\mathbb{P}^{t,\theta,\nu}[X_{[d]}] \in \Gamma^{t,\nu}$$

where for some backward map $B^{t,\nu}$

$$\theta = B^{t,\nu}(\mu).$$

Proof. By Lem. 7 if

$$\mu[L] \in (\mathcal{M}_{t,\nu})^\circ,$$

there is at least one weight parameter $\theta[L] \in \mathbb{R}^p$ with

$$\langle \mathbb{P}^{t,\theta,\nu}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} = \mu[L]$$

and $B^{t,\nu}(\mu)$ with respect to any backward map $B^{t,\nu}$ is such one. With Lem. 8 we conclude that $\mathbb{P}^{(t,B^{t,\nu}(\mu),\nu)}$ is the maximum entropy distribution with respect to the moment constraint on t by $\mu[L]$ and the base measure ν . $\square$

Note that while the maximum entropy distribution is unique, the weight vector θ representing the maximum entropy distribution is not always unique. This is reflected by multiple possible choices of backward mappings. Only if the exponential family is minimal (see Wainwright and Jordan (2008)) then the weight vector is unique. However, we did not restrict to such situations, since exponential families on refined base measures are not minimal.

4 Generic maximum entropy distributions

In the previous section we have shown that the members of an exponential family are maximum entropy distribution. However, their mean parameters build the relative interior of the mean polytope, but are never on faces of smaller dimension. We here characterize the maximum entropy distributions with respect to mean parameters on the full mean polytope.

4.1 Main result

Our main result generalizes the maximum entropy characterization of Thm. 3 to arbitrary mean parameters.

Theorem 4 (Generic characterization of Maximum Entropy Solutions). *Let t be a statistic and ν a base measure. For any $\theta[L]$ and $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ the distribution*

$$\mathbb{P}^{(t,\theta,\nu^{t,\mathcal{F}})}[X_{[d]}]$$

is a maximum entropy distribution.

For any $\mu[L]$ the maximum entropy problem has a feasible distribution, if and only if $\mu[L] \in \mathcal{M}_{t,\nu}$. If $\mu[L] \in \mathcal{M}_{t,\nu}$ the maximum entropy distribution is $\mathbb{P}^{(t,\theta,\nu^{t,\mathcal{F}})}$ where $\mathcal{F} = \mathcal{F}(\mu)$ and

$$\theta = B^{t,\nu^{t,\mathcal{F}}}(\mu)$$

where $\nu^{t,\mathcal{F}}$ is the refined base measure (see Def. 3) and $B^{t,\nu^{t,\mathcal{F}}}$ is any backward map to the exponential family $\Gamma^{t,\nu^{t,\mathcal{F}}}$ (see Def. 8).

For any hypergraph $\mathcal{G}$ and any face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ we have that all maximum entropy distributions to mean parameters on $\mathcal{F}^\circ$ are in $\Lambda^{t,\mathcal{G},\nu}$ if $\mathcal{F}$ is representable with respect to $\mathcal{G}$ (see Def. 4).

To prepare for the proof of this theorem we first show in an auxiliary lemma that we can reduce the set of feasible distributions in Problem $P_{t,\mu,\nu}$.

Lemma 9. *For any $\mu[L] \in \mathcal{M}_{t,\nu}$ and a face $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ with $\mu[L] \in (\mathcal{F})^\circ$ we have that the solutions of $P_{t,\mu,\nu}$ and $P_{t,\mu,\nu^{t,\mathcal{F}}}$ coincide.*

Proof. By Lem. 3 all feasible distributions are representable by the with the face measure refined base measure. We have that any for $P_{t,\mu,\nu}$ feasible distribution $\mathbb{P}[X_{[d]}]$ satisfies

$$\langle \mathbb{P}[X_{[d]}], \nu^{t,\mathcal{F}} \rangle_{[\emptyset]} = 1$$

and thus $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$. Conversely, any $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$ satisfies

$$\langle \mathbb{P}[X_{[d]}], \nu^{t,\mathcal{F}} \rangle_{[\emptyset]} = \langle \mathbb{P}[X_{[d]}], \nu \rangle_{[\emptyset]} = 1$$

and thus $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu}$. Problem $P_{t,\mu,\nu}$ is thus equal to

$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}} \mathbb{H}_\nu[\mathbb{P}[X_{[d]}]] \quad \text{subject to} \quad \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L], \nu[X_{[d]}] \rangle_{[L]} = \mu^*[L]$$

We further have that any $\mathbb{P}[X_{[d]}] \in \Lambda^{t,\nu^{t,\mathcal{F}}}$

$$\mathbb{H}_\nu[\mathbb{P}[X_{[d]}]] = \mathbb{H}_{\nu^{t,\mathcal{F}}}[\mathbb{P}[X_{[d]}]]$$

and arrive together with the above equivalence at the claim. $\square$

Proof of Thm. 4. To show that for any $\theta[L]$ and $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ the distribution $\mathbb{P}^{(t,\theta,\nu^{t,\mathcal{F}})}$ is a maximum entropy distribution, we apply Lem. 9 and Thm. 3.

If $\mu[L] \notin \mathcal{M}_{t,\nu}$ then Problem $P_{t,\mu,\nu}$ is not feasible by definition. In case $\mu[L] \in \mathcal{M}_{t,\nu}$ we find $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ with $\mathcal{F} = \mathcal{F}(\mu)$ and by Lem. 9 the solution of maximum entropy problem Problem $P_{t,\mu,\nu}$ coincides with $P_{t,\mu,\nu^{t,\mathcal{F}}}$. Since by Lem. 2 the face is the polytope

$$\mathcal{F} = \mathcal{M}_{t,\nu^{t,\mathcal{F}}}$$

we find a weight parameter

$$\theta = B^{t,\nu^{t,\mathcal{F}}}(\mu).$$

and by Thm. 3 $\mathbb{P}^{(t,\theta,\nu^{t,\mathcal{F}})}$ is the corresponding maximum entropy distribution.

To show the last claim, it is enough to show that for any by $\mathcal{G}$ representable face $\mathcal{F}$ and for any $\theta \in \mathbb{R}^p$ we have

$$\mathbb{P}^{(t,\theta,\nu^{t,\mathcal{F}})} \in \Lambda^{t,\mathcal{G},\nu}.$$

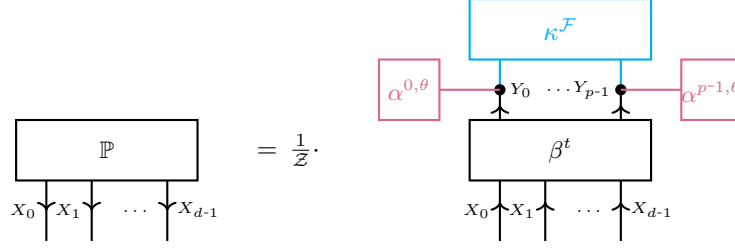


Figure 2: Tensor network decomposition of maximum entropy distributions to the constraint $\mu[L] = \langle \mathbb{P}[X_{[d]}], \sigma^t[X_{[d]}, L] \rangle_{[L]}$. Blue: Boolean tensor $\kappa^{\mathcal{F}}[Y_{[p]}]$ dependent on the face $\mathcal{F}(\cdot)(\mu)$, which represents the refinement of the base measure to the face, and is possibly non-elementary. Red: Elementary activation tensor $\alpha^\theta[Y_{[p]}] = \bigotimes_{s \in [n]} \alpha^{\ell, \theta}[Y_\ell]$ to the weight $\theta[L] = B^{t, \nu^{t, \mathcal{F}}}(\mu)$.

To this end, we construct activation cores as follows. By Def. 4 and Lem. 4 there is a tensor network $\kappa^{\mathcal{G}}$ such that

$$\nu^{t, \mathcal{F}}[X_{[d]}] = \langle \kappa^{\mathcal{G}} \cup \{\nu[X_{[d]}]\} \rangle_{[X_{[d]}]}.$$

For each $\ell \in [p]$ we find an edge $e(\ell) \in \mathcal{E}$ with $\ell \in e(\ell)$. Now we define a tensor network $\tau^{\mathcal{G}}$ by the cores to $e \in \mathcal{E}$ as

$$\tau^e[X_e] = \langle \{\kappa^e[X_e]\} \cup \{\alpha^{\ell, \theta}[Y_\ell] : e = e(\ell)\} \rangle_{[X_e]}.$$

Then, we have

$$\langle \beta^t[Y_{[p]}, X_{[d]}], \langle \tau^{\mathcal{G}} \rangle_{[Y_{[p]}]} \rangle_{[X_{[d]}]} = \langle \beta^t[Y_{[p]}, X_{[d]}], \alpha^\theta[Y_{[p]}], \langle \kappa^{\mathcal{G}} \rangle_{[Y_{[p]}]} \rangle_{[X_{[d]}]}$$

and therefore

$$\mathbb{P}^{(t, \theta, \nu^{t, \mathcal{F}})} = \frac{\langle \{\beta^t[Y_{[p]}, X_{[d]}]\} \cup \tau^{\mathcal{G}} \rangle_{[X_{[d]}]}}{\langle \{\beta^t[Y_{[p]}, X_{[d]}], \nu[X_{[d]}]\} \cup \tau^{\mathcal{G}} \rangle_{[\emptyset]}}.$$

We conclude $\mathbb{P}^{(t, \theta, \nu^{t, \mathcal{F}})} \in \Lambda^{t, \mathcal{G}, \nu}$.

□

4.2 Family of maximum entropy distributions

The family of maximum entropy distributions is the set

$$\Gamma^{t, \nu} = \{\mathbb{P}[X_{[d]}] : \exists \mu \in \mathcal{M}_{t, \nu} : \mathbb{P}[X_{[d]}] \text{ is a solution of Problem } P_{t, \mu, \nu}\}.$$

Thm. 4 suggest a parametrization of this family as tuples of corresponding faces $\mathcal{F}$ and weight vectors $\mu[L]$. As sketched in Figure 2 the activation tensor of each maximum entropy distribution is decomposed into a possibly non-elementary boolean tensor corresponding to the base measure refinement on a face $\mathcal{F}$, and an elementary tensor dependent on the weight vector $\theta \in \mathbb{R}^p$. We thus define the canonical parameter space

$$\mathcal{P}_{t, \nu} := L(\mathcal{M}_{t, \nu}) \times \mathbb{R}^p$$

and have

$$\Gamma^{t, \nu} = \left\{ \frac{\langle \alpha^\theta[Y_{[p]}], \kappa^{\mathcal{F}}[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}] \rangle_{[X_{[d]}]}}{\langle \alpha^\theta[Y_{[p]}], \kappa^{\mathcal{F}}[Y_{[p]}], \beta^t[Y_{[p]}, X_{[d]}], \nu[X_{[d]}] \rangle_{[\emptyset]}} : \theta[L] \in \mathbb{R}^p, \mathcal{F} \in \mathcal{P}_{t, \nu} \right\}.$$

Note that these families are disjoint, since each member of an exponential family has support by the support of the face measure.

5 Boolean statistics

We say that a statistic is boolean, if $n_\ell = 2$ for all $\ell \in [p]$. The mean polytope is then a 0/1-polytope Ziegler (2000), which vertices are exactly the contained boolean vectors. The special case of boolean statistics is especially interesting for two reasons. First, we can provide additional intuition on those faces representable by elementary face activating tensors, namely based on intersections with cube faces. Based on these properties we also show that any elementary Computation-Activation Network to a boolean statistic is a maximum entropy distribution. Second, the statistic itself can be interpreted in terms of logical formulas. We exploit this property to provide examples and to characterize each 0/1-polytope as a mean polytope to a collection of propositional formulas.

5.1 Characterization of elementary representable faces

Orienting on Example 3 we define cube-likeness of faces and polytopes.

Definition 9. We say that a face $\mathcal{F}$ of $\mathcal{M}_{t,\nu}$ is cube-like, if it is empty or there is $\mathcal{U} \subset [p]$ and $y_{\mathcal{U}} \in \times_{\ell \in \mathcal{U}} [2]$ such that

$$\mathcal{F} = \mathcal{M}_{t,\nu} \cap \mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}.$$

Here we denote by $\mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}$ the faces of the hypercube $[0, 1]^p$ (see Example 3). We further say that a polytope $\mathcal{M}_{t,\nu}$ is cube-like, if all faces $\mathcal{F} \in L(\mathcal{M}_{t,\nu})$ are cube-like.

We now show that a face is cube-like if and only if it is representable by an elementary tensor.

Theorem 5. Let t be a boolean statistic, ν a base measure and $\mathcal{F}$ be a face of $\mathcal{M}_{t,\nu}$. Then the following are equivalent:

- (i) $\mathcal{F}$ is cube-like (see Def. 9).
- (ii) $\mathcal{F}$ is representable by an elementary tensor (see Def. 4).

Proof. If the face is empty, i.e. $\mathcal{F} = \emptyset$, it is by definition cube-like and has $0 [Y_{[p]}]$ as an elementary activation tensor. We therefore assume in the following $\mathcal{F} \neq \emptyset$.

(i) $\Rightarrow$ (ii): Let us assume that $\mathcal{F}$ is cube-like, that is there is $\mathcal{U} \subset [p]$ and $y_{\mathcal{U}} \in \times_{\ell \in \mathcal{U}} [2]$ such that $\mathcal{F} = \mathcal{M}_{t,\nu} \cap \mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}$. We use that $\mathcal{F}$, $\mathcal{M}_{t,\nu}$ and $\mathcal{F}^{(\mathcal{U}, y_{\mathcal{U}})}$ are the convex hulls of the cube vertex sets $\mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}}$, $\mathcal{N}_{t,\mathcal{M}}$ and

$$\mathcal{N}^{\mathcal{U}, y_{\mathcal{U}}} = \{v[L] : \forall \ell \in \mathcal{U} : v[L = \ell] = y_{\mathcal{U}}\}, \quad (1)$$

which implies that

$$\mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}} = \mathcal{N}^{\mathcal{U}, y_{\mathcal{U}}} \cap \mathcal{N}_{t,\mathcal{M}}.$$

Thus, we can choose $\mathcal{U} = \mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}}$ for the representation of the face $\mathcal{F}$ (see Def. 4) and further have that

$$\sum_{v[L] \in \mathcal{N}^{(\mathcal{U}, y_{\mathcal{U}})}} \epsilon_{v[L]} [Y_{[p]}] = \bigotimes_{\ell \in [p]} \kappa^{\ell, (\mathcal{U}, y_{\mathcal{U}})} [Y_{\ell}].$$

We have thus found an elementary activation tensor for the face $\mathcal{F}$.

(ii) $\rightarrow$ (i) Conversely, let $\xi [Y_{[p]}]$ be an elementary activation tensor of the face $\mathcal{F}$ in $\mathcal{M}_{t,\nu}$. Since $\xi [Y_{[p]}]$ is the sum of different one-hot encodings it is boolean and we find an elementary decomposition $\xi [Y_{[p]}] = \bigotimes_{\ell \in [p]} \xi^{\ell} [Y_{\ell}]$ such that the leg vectors $\xi^{\ell} [Y_{\ell}]$ are boolean. Since $\mathcal{F} \neq \emptyset$ we further have for $\ell \in [p]$ that $\xi^{\ell} [Y_{\ell}] \neq 0 [Y_{\ell}]$, and thus $\xi^{\ell} [Y_{\ell}] \in \{\epsilon_0 [Y_{\ell}], \epsilon_1 [Y_{\ell}], \mathbb{I} [Y_{\ell}]\}$. We construct a set $\mathcal{U} \subset [p]$

$$\mathcal{U} = \{\ell : \xi^{\ell} [Y_{\ell}] \neq \mathbb{I} [Y_{\ell}]\}$$

and a tuple

$$y_{\ell} = \begin{cases} 1 & \text{if } \xi^{\ell} [Y_{\ell}] = \epsilon_1 [Y_{\ell}] \\ 0 & \text{if } \xi^{\ell} [Y_{\ell}] = \epsilon_0 [Y_{\ell}] \end{cases}.$$

By construction $\xi^{\ell} [Y_{\ell}] = \kappa^{\ell, (\mathcal{U}, y_{\mathcal{U}})} [Y_{\ell}]$ (see Example 3) follows for all $\ell \in [p]$ and therefore $\xi [Y_{[p]}] = \kappa^{(\mathcal{U}, y_{\mathcal{U}})} [Y_{[p]}]$. We therefore have for the set (1)

$$\xi [Y_{[p]}] = \sum_{v \in \mathcal{N}^{\mathcal{U}, y_{\mathcal{U}}}} \epsilon_v [Y_{[p]}]$$

and since $\xi [Y_{[p]}]$ is an activation tensor for $\mathcal{F}$, that

$$\mathcal{N}_{t,\mathcal{M}}^{\mathcal{F}} = \mathcal{N}^{\mathcal{U},y\mathcal{U}} \cap \mathcal{N}_{t,\mathcal{M}}.$$

Taking convex hulls on both sides, this is equivalent to

$$\mathcal{F} = \mathcal{F}^{(\mathcal{U},y\mathcal{U})} \cap \mathcal{M}_{t,\nu}.$$

We conclude that $\mathcal{F}$ is cube-like. $\square$

Thm. 5 implies that the maximum entropy distributions to hypercubes (see Example 3) and simplices (see Example 4) are in $\Lambda^{t,\text{EL},\nu}$, since these polytopes are cube-like.

We can further show, that for boolean statistics, any distribution in $\Lambda^{t,\text{EL},\nu}$ is a maximum entropy distribution.

Theorem 6. *Let t be a boolean statistic. Any distribution in $\Lambda^{t,\text{EL},\nu}$ is a maximum entropy distribution with respect to (t, μ, ν) where μ is its mean parameter.*

Proof. Let $\mathbb{P} \in \Lambda^{t,\text{EL},\nu}$ be arbitrary. We then find an elementary tensor $\bigotimes_{s \in [n]} \tau^\ell [Y_\ell]$ and $\mathcal{Z} \in \mathbb{R}$ such that

$$\mathbb{P} [X_{[d]}] = \frac{1}{\mathcal{Z}} \left\langle \beta^t [Y_{[p]}, X_{[d]}], \bigotimes_{s \in [n]} \tau^\ell [Y_\ell] \right\rangle_{[X_{[d]}]}.$$

We then define for each $\ell \in [p]$ the vector $\kappa^\ell [Y_\ell]$ as the support of $\tau^\ell [Y_\ell]$ and

$$\theta [L = \ell] = \begin{cases} \ln \left[\frac{\tau^\ell [Y_\ell = 1]}{\tau^\ell [Y_\ell = 0]} \right] & \text{if } \tau^\ell [Y_\ell = 0], \tau^\ell [Y_\ell = 1] \neq 0 \\ 0 & \text{else} \end{cases}.$$

Since t is a boolean statistic the $\bigotimes_{\ell \in [p]} \kappa^\ell [Y_\ell]$ is a face activation tensor to the face

$$\mathcal{F} := \mathcal{M}_{t,\mathbb{I}} \cap \mathcal{F}^{(\mathcal{U},y\mathcal{U})}$$

where $\mathcal{U} = \{\ell : \kappa^\ell [Y_\ell] \neq \mathbb{I} [Y_\ell]\}$ and for $\ell \in \mathcal{U}$

$$y_\ell = \begin{cases} 0 & \text{if } \kappa^\ell [Y_\ell = 1] = 0 \\ 1 & \text{else} \end{cases}.$$

Therefore, $\mathbb{P}$ is the maximum entropy distribution to the canonical parameters $\mathcal{F}$ and θ . $\square$

5.2 Interpretation by propositional formulas

We can understand each feature as a propositional formula and the variables $X_{[d]}$ as atoms (possibly after a binarization). We use propositional connectives for syntactical representations of the formulas and denote by $\neg^1$ the logical negation and by $\neg^0$ the identity. A boolean tensor $f [X_{[d]}]$ is called satisfiable, if there is at least one $x_{[d]} \in \times_{k \in [d]} [m_k]$ with $f [X_{[d]} = x_{[d]}] = 1$.

Theorem 7. *Let t be a boolean statistic. Any boolean vector $v [L] \in \{0, 1\}^p$ is a vertex of $\mathcal{M}_{t,\mathbb{I}}$ if and only if*

$$f_v [X_{[d]}] := \bigwedge_{\ell \in [p]} \neg^{1-v[L=\ell]} f_\ell [X_{[d]}]$$

is satisfiable.

Proof. Let $v [L] \in \{0, 1\}^p$ be arbitrary. If and only if $v [L]$ is a vertex of $\mathcal{M}_{t,\mathbb{I}}$ we find $x_{[d]} \in \times_{k \in [d]} [m_k]$ with $v [L] = \sigma^t X_{[d]} = x_{[d]}, L$. This is further equivalent to $f_\ell [X_{[d]} = x_{[d]}] = v [L = \ell]$ for any $\ell \in [p]$, $\neg^{1-v[L=\ell]} f_\ell [X_{[d]} = x_{[d]}] = 1$ for any $\ell \in [p]$, and $f_v [X_{[d]} = x_{[d]}] = 1$. $\square$

Since the presence of boolean vectors in $\mathcal{M}_{\ell, \mathbb{I}}$ is by Thm. 7 related to the satisfiability of the conjunction $f_v [X_{[d]}]$, the absence is related to entailment statements. To be more precise, for any $v [L] \in \{0, 1\}^p$ and arbitrary $\mathcal{U} \subset [p]$ we have $v [L] \notin \mathcal{M}_{\ell, \mathbb{I}}$, if and only if

$$\left(\bigwedge_{\ell \in \mathcal{U}} \neg^{1-v[L=\ell]} f_\ell [X_{[d]}] \right) \models \left(\bigvee_{\ell \notin \mathcal{U}} \neg^{v[L=\ell]} f_\ell [X_{[d]}] \right).$$

By definition (see Russell and Norvig (2021)) the latter holds if and only if

$$\left(\bigwedge_{\ell \in \mathcal{U}} \neg^{1-v[L=\ell]} f_\ell [X_{[d]}] \right) \wedge \neg \left(\bigvee_{\ell \notin \mathcal{U}} \neg^{v[L=\ell]} f_\ell [X_{[d]}] \right) = f_v [X_{[d]}]$$

is not satisfiable.

Atomic formulas and minterm formulas correspond to hypercubes and simplices as we show next.

Example 5 (Atomic formulas). *Let us consider the case of atomic formulas $f_\ell [X_{[d]}]$ defined with the coordinates at $x_{[d]} \in \times_{k \in [d]} [m_k]$ by*

$$f_\ell [X_{[d]} = x_{[d]}] = x_\ell.$$

The mean polytope in this case is the d -dimensional hypercube (see Example 3)

$$\mathcal{M}_{\mathcal{F}_{[d]}, \mathbb{I}} = [0, 1]^p$$

which is called a simple polytope, since each vertex is contained in the minimal number of d facets.

Each face is characterized by the projections onto each variable, which is either $\{0\}$, $\{1\}$ or $[0, 1]$. The projections are represented by the tuple $(\mathcal{U}, y_\mathcal{U})$ defined in the following way:

- *We define the set $\mathcal{U} \subset [d]$ of variables, such that the projection onto the variable is $\{0\}$ or $\{1\}$*
- *We define to each $\ell \in \mathcal{U}$ an index $y_\ell = 0$ if the projection is $\{0\}$ and $y_\ell = 1$ if the projection is $\{1\}$.*

Trivially, each face of the hypercube is a cube face.

More general, the mean polytope to a collection $f_\ell [X_{[d]}]$ of propositional formulas is the hypercube, if and only if for any boolean vector $v [L]$ we have

$$\left\langle \bigwedge_{\ell \in [p]} \neg^{1-v[L=\ell]} f_\ell [X_{[d]}] \right\rangle_{[\emptyset]} \neq 0.$$

This means that no conjunction of possibly negated formulas from this statistic entails nor contradicts another formula.

Example 6 (Minterm formulas build simplices). *The set of minterm formulas is indexed by $v [L] \in \{0, 1\}^p$ and given by*

$$f_v [X_{[d]}] = \bigwedge_{\ell \in [d]} \neg^{1-v[X_{[d]}=x_{[d]}]} X_\ell = \epsilon_{v[L]} [X_{[d]}]$$

where by X_ℓ we denote the ℓ -th atomic formula (see Example 6). The mean polytope to the minterm statistic and base measure ν is the standard simplex of dimension

$$|\{v [L] : v \in \{0, 1\}^p, \nu [X_{[d]} = v [L]] \neq 0\}| - 1,$$

that is the set

$$\mathcal{M}_{\mathcal{F}_\wedge, \nu} = \text{conv} (\epsilon_{v[L]} [Y_{[p]}] : v \in \{0, 1\}^p, \nu [X_{[d]} = v [L]] \neq 0).$$

We notice that for boolean base measures $\mathcal{M}_{\mathcal{F}_\wedge, \nu} = \Delta^\nu$. The simplex is further cube-like, since each face is indexed by a subset $\mathcal{U} \subset \{0, 1\}^p$ of vertices and the intersection of the simplex with the cube face $\mathcal{F}^{\{0, 1\}^p \setminus \mathcal{U}, 1_{\{0, 1\}^p \setminus \mathcal{U}}}$.

More general the mean polytope to a boolean statistic is a standard simplex, if and only if each target coordinate restriction $f_{\tilde{\ell}}$ contradicts all others, that is

$$\forall \ell \neq \tilde{\ell} \in [p] : f_\ell \models \neg f_{\tilde{\ell}}.$$

Let us now provide examples of statistics, which polytopes are not cube-like.

Example 7 (Maximum entropy distribution with non-elementary activation cores). *Consider two atomic variables X_0 and X_1 and a statistic $\mathcal{F}$ consisting in the formulas*

$$f_0 [X_0, X_1] = (X_0 \wedge X_1) \quad , \quad f_1 [X_0, X_1] = (X_0 \Rightarrow X_1)$$

with the coordinatewise expressions

$$f_0 [X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] \quad , \quad f_1 [X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} [X_0, X_1] .$$

We can think of X_0 as a feature on an invoice, and X_1 as a feature on the accounting proposal.

From this we have

$$\begin{aligned} \beta^{(f_0, f_1)} [Y_0 = 0, Y_1 = 0, X_0, X_1] &= \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} [X_0, X_1] \quad , \quad \beta^{(f_0, f_1)} [Y_0 = 0, Y_1 = 1, X_0, X_1] = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} [X_0, X_1] \quad , \\ \beta^{(f_0, f_1)} [Y_0 = 1, Y_1 = 0, X_0, X_1] &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} [X_0, X_1] \quad \text{and} \quad \beta^{(f_0, f_1)} [Y_0 = 1, Y_1 = 1, X_0, X_1] = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} [X_0, X_1] . \end{aligned}$$

Since the only vanishing slice of $\beta^{\mathcal{F}}$ with respect to the head variables is that to $y_{0,1} = (1, 0)$, the vertices of the mean polytope are the vectors to the other head indices. The mean polytope is the convex hull of these vertices

$$\mathcal{M}_{(f_0, f_1)} = \text{conv} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} [L], \begin{pmatrix} 0 \\ 1 \end{pmatrix} [L], \begin{pmatrix} 1 \\ 1 \end{pmatrix} [L] \right) .$$

This polytope has a non cube-like face (sketched blue in Figure 3), which is the convex hull of the vertices $[0 0]^T$, $[1 1]^T$. This face is parametrized by the (CP-rank 2) hard activation core

$$\kappa^{(0,0),(1,1)} [Y_0, Y_1] = \epsilon_{(0,0)} [Y_0, Y_1] + \epsilon_{(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} [Y_0, Y_1]$$

and has the face measure

$$\left\langle \kappa^{(0,0),(1,1)} [Y_0, Y_1], \beta^{\mathcal{F}} [Y_0, Y_1, X_0, X_1] \right\rangle_{[X_0, X_1]} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} [X_0, X_1] .$$

Any mean parameter μ on the interior of that face can be parametrized by a scalar $\lambda \in (0, 1)$

$$\mu_\lambda [L] = (\lambda \quad \lambda) [L] .$$

With the canonical parameters $\theta [L] \in \mathbb{R}^2$ of the maximum entropy distributions on this face by

$$\mathbb{P} [X_0, X_1] = \frac{1}{1 + \exp [\theta [L = 0] + \theta [L = 1]]} \begin{pmatrix} 0 & 0 \\ 1 & \exp [\theta [L = 0] + \theta [L = 1]] \end{pmatrix} [X_0, X_1]$$

we get the correspondence by the sigmoid

$$\lambda = \frac{1}{1 + \exp [-(\theta [L = 0] + \theta [L = 1])] .$$

Note, that the hard activation core $\kappa^{(0,0),(1,1)} [Y_0, Y_1]$ to the blue face is the only non-elementary activation core. While the vertices have always elementary cores, the further non-vertex faces have elementary activation cores

$$\kappa^{(0,0),(1,0),(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} [Y_0, Y_1] = \mathbb{I} [Y_0] \otimes \mathbb{I} [Y_1] , \quad \kappa^{(0,0),(1,0)} [Y_0, Y_1] = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} [Y_0, Y_1] = \mathbb{I} [Y_0] \otimes \epsilon_0 [Y_1] ,$$

$$\kappa^{(1,0),(1,1)} [Y_0, Y_1] = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} [Y_0, Y_1] = \epsilon_0 [Y_0] \otimes \mathbb{I} [Y_1] .$$

The maximum entropy distributions to mean parameters on the interior of all other faces than the blue face are represented by Computation-Activation Networks with only elementary activation cores.

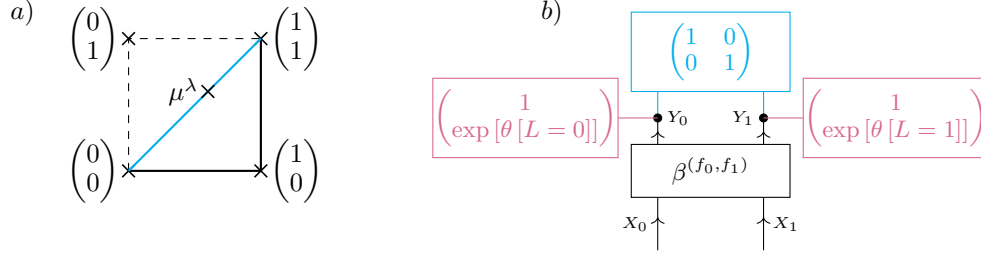


Figure 3: a) Mean polytope of the statistic $\mathcal{F} = (X_0 \wedge X_1, X_0 \Rightarrow X_1)$ (thick), as a subset of the cube $[0, 1]^2$ (dashed). The blue line is the face of the polytope, which is not cube like, that is not an intersection of the polytope with the faces of the polytope. We further define for $\lambda \in (0, 1)$ a mean parameter $\mu_\lambda[L] = [\lambda \lambda]^T$ which is on the interior of the blue face. b) Corresponding Computation-Activation Network being the maximum entropy distribution reproducing $\mu_\lambda[L]$, when λ is the sigmoid of $\theta[L = 0] + \theta[L = 1]$.

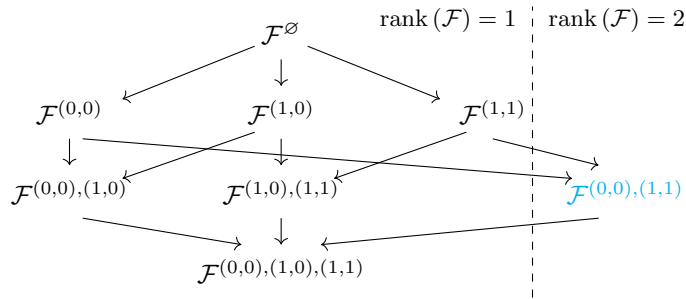


Figure 4: Face lattice $L((X_0 \wedge X_1, X_0 \Rightarrow X_1))$ to Example 5. The directed arrows represent inclusion of the faces, which is a partial order of the faces. The face $\mathcal{F}^{(0,0),(1,1)}$ is the only face, which is not representable by an elementary hypergraph. It is representable in a CP with hidden rank 2

Example 8 (TT representation of diagonal faces). *Consider the vertex set*

$$\mathcal{N} := \{0, 1\}^p / v[L]$$

for some boolean vector $v[L]$. We are interested in the face $\mathcal{F}^{\triangleleft, 1}$ with the normal $\frac{1}{2}(2v[L] - \mathbb{I}[L])$. Its vertices are the p elements of $\{0, 1\}^p$, which differ from $v[L]$ in exactly one coordinate (i.e. those with Hamming distance of 1 from $v[L]$). We denote these vertices by v^ℓ for $\ell \in [p]$, which coordinates to $\tilde{\ell} \in [p]$ are

$$v^\ell[L = \tilde{\ell}] = \begin{cases} v[L = \tilde{\ell}] & \text{if } \tilde{\ell} \neq \ell \\ 1 - v[L = \tilde{\ell}] & \text{if } \tilde{\ell} = \ell \end{cases}.$$

We now represent their sum in an TT with hidden ranks $r_0 = \dots = r_{p-2} = 2$. This format describes tensor networks on a hypergraph

$$\text{TT} = (X_{[d]} \cup I_{[d-1]}, \{\{I_{k-1}, X_k, I_k\} : k \in \{1, \dots, d-2\}\} \cup \{\{X_0, I_0\}, \{I_{d-2}, X_{d-1}\}\}).$$

The boolean hidden variables are denoted by $I_{[p-1]}$ and can be interpreted as indicators, whether the coordinate flip has happened in $[\ell]$ coordinates. We now construct a TT cores for $\ell = 0$ by

$$\tau^0[Y_0, I_0] = \epsilon_{v[L=0]}[Y_0] \otimes \epsilon_0[I_0] + \epsilon_{1-v[L=0]}[Y_0] \otimes \epsilon_1[I_0]$$

further for $\ell \notin \{0, p-1\}$ the cores

$$\begin{aligned} \tau^\ell[I_{\ell-1}, Y_\ell, I_\ell] &= \epsilon_0[I_{\ell-1}] \otimes \epsilon_{1-v[L=\ell]}[Y_\ell] \otimes \epsilon_1[I_\ell] + \epsilon_0[I_{\ell-1}] \otimes \epsilon_{v[L=\ell]}[Y_\ell] \otimes \epsilon_0[I_\ell] \\ &\quad + \epsilon_1[I_{\ell-1}] \otimes \epsilon_{v[L=\ell]}[Y_\ell] \otimes \epsilon_1[I_\ell] \end{aligned}$$

and for $\ell = p-1$

$$\tau^{p-1}[I_{p-1}, Y_{p-1}] = \epsilon_0[I_{p-1}] \otimes \epsilon_{1-v[L=p-1]}[Y_{p-1}] + \epsilon_1[I_{p-1}] \otimes \epsilon_{v[L=p-1]}[Y_{p-1}].$$

For this tensor network in the TT format we have

$$\sum_{\ell \in [p]} \epsilon_{v^\ell} [Y_{[p]}] = \langle \{ \tau^0 [Y_0, I_0], \tau^{p-1} [I_{p-1}, Y_{p-1}] \} \cup \{ \tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] : \ell \notin \{0, p-1\} \} \rangle_{[Y_{[p]}]} .$$

The TT multirank of 2 is furthermore minimal, since each matrification based on a partition of $[p]$ into non-empty sets has a matrix rank of 2. With respect to such partitions also the tensor $\epsilon_v [Y_{[p]}] + \left(\sum_{\ell \in [p]} \epsilon_{v^\ell} [Y_{[p]}] \right)$ has matrix ranks of 2. Since these are the only two activation tensors for the face $\mathcal{F}^{\triangleleft, 1}$.

Example 9 (Generalization of Example 8 to larger TT ranks). We now generalize the construction of Example 8 by using the Hadamard distance $d(\cdot, \cdot)$ in $\{0, 1\}^p$, which counts the number of coordinates two vertices differ in. For $s \in \{1, \dots, p-1\}$ we define for a fixed $v [L] \in \{0, 1\}^p$ a polytope as the convex hull

$$\{\tilde{v}[L] : d(\tilde{v}, v) \geq s\} .$$

The face to the normal $\frac{1}{2}(2v [L] - \mathbb{I} [L])$ is the convex hull

$$\mathcal{F}^{\triangleleft, s} := \text{conv}(\{\tilde{v}[L] : d(\tilde{v}, v) = s\})$$

containing $\binom{p}{s}$ vertices. We label these vertices by subsets $\mathcal{U} \subset [p]$ of cardinality s and define for $\ell \in [p]$

$$v^\mathcal{U} [L = \ell] = \begin{cases} v [\ell \in [p]] & \text{if } \ell \notin \mathcal{U} \\ 1 - v [\ell \in [p]] & \text{if } \ell \in \mathcal{U} \end{cases}$$

We now construct a TT with hidden variables I_ℓ and ranks $r_\ell = \min(\ell + 1, p - \ell + 1, s + 1)$ to represent the sum of their one-hot encodings. To this end, let there be TT cores for $\ell = 0$ by

$$\tau^0 [Y_0, I_0] = \epsilon_{v[L=0]} [Y_0] \otimes \epsilon_0 [I_0] + \epsilon_{1-v[L=0]} [Y_0] \otimes \epsilon_1 [I_0]$$

further for $\ell \notin \{0, p-1\}$ the cores

$$\tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] = \sum_{\ell \in \{\max(s-p+\ell, 0), \dots, \max(\ell, s)\}} \epsilon_\ell [I_\ell] \otimes (\epsilon_{v[L=\ell]} [Y_\ell] \otimes \epsilon_\ell [I_{\ell+1}] + \epsilon_{1-v[L=\ell]} [Y_\ell] \otimes \epsilon_{\ell+1} [I_{\ell+1}])$$

and for $\ell = p-1$

$$\tau^{p-1} [I_{p-1}, Y_{p-1}] = \epsilon_0 [I_{p-1}] \otimes \epsilon_{1-v[L=p-1]} [Y_{p-1}] + \epsilon_1 [I_{p-1}] \otimes \epsilon_{v[L=p-1]} [Y_{p-1}] .$$

Based on the interpretation, that the hidden variables I_ℓ count the Hamming distance of the vectors $v [L]|_{\mathbb{R}^\ell \times 0_{p-\ell}}$ and the respective $v^\mathcal{U} [L]|_{\mathbb{R}^\ell \times 0_{p-\ell}}$ one can show that

$$\sum_{\mathcal{U} \subset [p] : |\mathcal{U}|=s} \epsilon_{v^\mathcal{U}} [Y_{[p]}] = \langle \{ \tau^0 [Y_0, I_0], \tau^{p-1} [I_{p-1}, Y_{p-1}] \} \cup \{ \tau^\ell [I_{\ell-1}, Y_\ell, I_\ell] : \ell \notin \{0, p-1\} \} \rangle_{[Y_{[p]}]} .$$

Example 10 (Generalization of Example 9 to arbitrary HT formats). Instead of aligning the Hamming count variables linearly, one can find a representation of the activation tensor

$$\sum_{\mathcal{U} \subset [p] : |\mathcal{U}|=s} \epsilon_{v^\mathcal{U}} [Y_{[p]}]$$

in an arbitrary directed acyclic tree hypergraph format $\mathcal{G}$, which hidden ranks are bounded by the number of leafs in the subtree and $s + 1$.

At any leaf of the tree we define a 2×2 matrix

$$\tau^\ell [Y_\ell, I_\ell] = \epsilon_{v[L=\ell]} [Y_\ell] \otimes \epsilon_0 [I_\ell] + \epsilon_{1-v[L=\ell]} [Y_\ell] \otimes \epsilon_1 [I_\ell] .$$

At each intermediate non-root hyperedge we choose an outgoing counting variable $I_{e^{\text{out}}}$ with dimension by $n_{e^{\text{out}}} = \min(\sum_{v \in e^{\text{in}}} n_{e^{\text{in}}}, s + 1)$ and define a tensor with the slices

$$\tau^e [I_{e^{\text{out}}}, I_{e^{\text{in}}}] = \beta^+ [Y_+ = i_{e^{\text{out}}}, I_{e^{\text{in}}}] .$$

We further build at the root hyperedge e a tensor

$$\tau^e [I_e] = \sum_{i : \langle i \rangle_{[\emptyset]} = s} \epsilon_{i_e} [I_e] .$$

Using the counting variable interpretation one can now show that

$$\sum_{\mathcal{U} \subset [p] : |\mathcal{U}|=s} \epsilon_{v^\mathcal{U}} [Y_{[p]}] = \langle \{ \tau^e [I_e] : e \in \mathcal{E} \} \cup \{ \tau^\ell [Y_\ell, I_\ell] : v \in \mathcal{V} \} \rangle_{[Y_{[p]}]} .$$

5.3 Construction of boolean statistics to polytopes

We now show, that any convex polytope with boolean vertices in $\mathbb{R}^p$ is the mean polytope to a boolean statistic. To construct the corresponding formulas we denote by $\mathcal{M}|_{\mathcal{U}}$ the restriction of the vectors in the set $\mathcal{M}$ to the coordinates in $\mathcal{U}$.

Theorem 8. *Let $\mathcal{M}$ an arbitrary polytope with boolean vertices in $\mathbb{R}^p$. Then we construct propositional formulas on atoms $X_{[p]}$ by*

$$f_0[X_0] = \begin{cases} \top & \text{if } \mathcal{M}|_{\{0\}} = \{1\} \\ \perp & \text{if } \mathcal{M}|_{\{0\}} = \{0\} \\ X_0 & \text{if } \mathcal{M}|_{\{0\}} = [0, 1] \end{cases}$$

and iteratively for $\ell \in [p]$ with $\ell \geq 1$ by

$$f_\ell[X_{[\ell+1]}] = \bigwedge_{v[L] \in (\mathcal{M} \cap \{0,1\}^p)|_{[\ell]}} \left(\left(\bigwedge_{\tilde{\ell} \in [\ell]} \neg^{1-v[L=\tilde{\ell}]} f_{\tilde{\ell}}[X_{[\tilde{\ell}+1]}] \right) \Rightarrow \begin{cases} \top & \text{if } (\mathcal{M} \cap v \times \mathbb{R}^{p-\ell})|_{\{\ell\}} = \{1\} \\ \perp & \text{if } (\mathcal{M} \cap v \times \mathbb{R}^{p-\ell})|_{\{\ell\}} = \{0\} \\ X_\ell & \text{if } (\mathcal{M} \cap v \times \mathbb{R}^{p-\ell})|_{\{\ell\}} = [0, 1] \end{cases} \right).$$

Here we denote by $\mathcal{M}|_V$ the projections of the vertices in $\mathcal{M}$ onto the subspaces V , and by 0_p the zero vector in $\mathbb{R}^p$.

Proof. We show per induction, that for any $\ell \in [p]$ the family of Hybrid Logic Networks with the statistic $f_{[\ell+1]}$ by the first $\ell + 1$ formulas has the mean polytope

$$\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}} = \mathcal{M}|_{[\ell]}. \quad (2)$$

" $\ell = 0$ ": The polytope $\mathcal{M}|_{[\ell]} = \{1\}$ (respectively $\mathcal{M}|_{[\ell]} = \{0\}$) is reproduced by f_0 being a tautology (respectively a contradiction). In the case $\mathcal{M}|_{[\ell]} = [0, 1]$ the polytope is reproduced by the any formula, which is neither a tautology nor a contradiction, and the atomic formula X_0 is an example of such an contingency. Since the projection of a 0-1 polytope onto the first coordinates is itself a 0-1 polytope, these are the only possible cases and we conclude that in all

$$\mathcal{M}_{f_{[\ell]}, \mathbb{I}} = \mathcal{M}|_{[\ell]}.$$

" $\ell \rightarrow \ell + 1$ ": We define for any $v \in \mathcal{M}_{f_{[\ell]}, \mathbb{I}} \cap \{0, 1\}^\ell$ the formula

$$f_{\ell, v}[X_{[\ell+1]}] := \bigwedge_{\tilde{\ell} \in [\ell]} \neg^{1-v[L=\tilde{\ell}]} f_{\tilde{\ell}}[X_{[\tilde{\ell}+1]}].$$

and notice that all formulas are satisfiable (since the corresponding vertices are in the $\mathcal{M}_{f_{[\ell]}, \mathbb{I}}$) and have disjoint models. We have for any boolean vector $v \in \{0, 1\}^\ell$ and $\lambda \in \{0, 1\}$ that $v \times \{\lambda\} \in \mathcal{M}|_{[\ell+1]}$, if and only if

$$v \in \mathcal{M}|_{[\ell]} \quad \text{and} \quad \lambda \in \mathcal{M} \cup v \times \mathbb{R}^{p-\ell}|_{[\ell]}.$$

With modus ponens on $f_{\ell+1}[X_{[\ell+2]}]$ this is further equivalent to the satisfaction of

$$f_{\ell, v}[X_{[\ell+1]}] \wedge \neg^{1-\lambda} f_{\ell+1}[X_{[\ell+2]}]$$

and thus to $v \times \{\lambda\} \in \mathcal{M}_{f_{[\ell+1]}, \mathbb{I}}$. The vertices of the 0/1 polytopes $\mathcal{M}_{f_{[\ell+1]}, \mathbb{I}}$ and $\mathcal{M}|_{[\ell+1]}$ thus coincide and both polytopes are therefore equivalent.

By induction the equation (2) holds for arbitrary $\ell \in [p]$. For $\ell = p - 1$ this is the claim. $\square$

5.4 Examples in statistical physics and coding theory

Example 11 (Ising model on 2 nodes). *Consider two boolean variables X_0, X_1 and the Ising statistic by three propositional formulas*

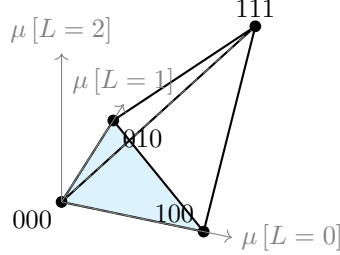
$$f_0[X_{[2]}] = X_0 \quad , \quad f_1[X_{[2]}] = X_1 \quad \text{and} \quad f_2[X_{[2]}] = X_0 \wedge X_1.$$

In the Ising interpretation, the boolean variables represent interacting spins at two locations. Their value is then measured by the first two formulas and their interaction by the third.

The vertices of the mean polytope to this statistic are

$$t(x_0, x_1) = [f_0[X_{[2]} = x_{[2]}], f_1[X_{[2]} = x_{[2]}], f_2[X_{[2]} = x_{[2]}]]^T = \begin{cases} [0, 0, 0]^T & \text{if } (x_0, x_1) = (0, 0) \\ [0, 1, 0]^T & \text{if } (x_0, x_1) = (0, 1) \\ [1, 0, 0]^T & \text{if } (x_0, x_1) = (1, 0) \\ [1, 1, 1]^T & \text{if } (x_0, x_1) = (1, 1) \end{cases}$$

and the mean polytope is the convex hull of these, sketched as:



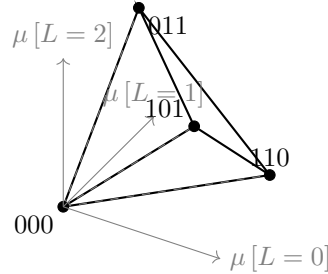
We notice, that the convex hull of any three out of the four vertices builds a facet. The only cube-like facet of these four is the convex hull of $\{[0, 0, 0]^T, [1, 0, 0]^T, [0, 1, 0]^T\}$ (sketched in cyan in the above plot) to which we have the parametrization by $(\mathcal{U}, y_{\mathcal{U}}) = (\{2\}, 0)$.

As a consequence, there are maximum entropy distributions of mean parameters in the Ising statistics, which do not have a representation by an elementary CompActNet in the Ising statistic.

Example 12 (Codeword Polytopes). Consider the codeword polytope with the vertices

$$\mathcal{N} = \left\{ y_{[3]} : \sum_{k \in [3]} y_k \mod 2 = 0 \right\} = \{(0, 0, 0), (0, 1, 1), (1, 0, 1), (1, 1, 0)\}$$

We interpret the vertices by the strings of length 3, which parity vanishes. Such constrained are common in the construction of codewords using parity-check codes (see Gallager (1963)).



All facets are non-cube-like, whereas all other faces are cube-like (the edges are labeled by the 6 facets of $[0, 1]^3$).

Each facet is characterized by the Hamming distance of exactly 1 to a single of the 4 cube vertices, which are not codewords (see Example 3.9 in Wainwright and Jordan (2008)). We can construct TT activation tensors with rank 2, which are calculating in the hidden ranks the Hamming distance to these vertex. Each core tensor in that format adds the contribution of a mode to the Hamming distance.

In more generality, Low-Density-Parity-Checking Codes (see Gallager (1963)) construct codewords by a collection of parity constraints on subsets of variables. Such polytopes are also known as cycle polytopes, and studied e.g. in Grötschel et al. (1993).

6 Outlook

This work characterized all discrete maximum entropy distributions as Computation-Activation Networks, a class of tensor networks. The mean polytope contains those mean parameters, to which entropy maximization problems are feasible. While exponential families are known to be the maximum entropy distributions to the interior of the

polytope, we showed that refining the base measure with respect to faces enables the representation of maximum entropy distributions to mean parameters on the particular face. To this end we utilized the tensor network formalism and showed how to represent exponential families and their generalizations to maximum entropy distributions.

One important application is estimation of maximum entropy distributions from data, such as learning in exponential families and graphical models Jordan (1998). Maximum likelihood problems are known to be dual to maximum entropy problems Koller and Friedman (2009) and therefore face similar representation challenges. Here the mean parameters of empirical distributions serve as sufficient statistics and are matched by trainable weights in algorithms such as iterative proportional fitting Darroch and Ratcliff (1972); Csiszar (1989). Optimizing parameters within a poor choice of learning architecture will lead to numerical instabilities, resulting from the absence of an optimum. Choosing the right tensor format representing the refined base measure guarantees the existence of an optimum.

Further research might derive a relation between tensor format complexity Hackbusch (2012) to notions of facet complexity Grötschel et al. (1993). The complexity of faces from randomly drawn polytopes is further interesting, especially in the asymptotic limit of large statistics Vershynin (2018); Wainwright (2019).

References

- Lawrence D. Brown. *Fundamentals of Statistical Exponential Families*. Institute of mathematical Statistics, Hayward, Calif, June 1987. ISBN 978-0-940600-10-2.
- Imre Csiszar. A Geometric Interpretation of Darroch and Ratcliff's Generalized Iterative Scaling. *The Annals of Statistics*, 17(3):1409–1413, September 1989. ISSN 0090-5364, 2168-8966. doi: 10.1214/aos/1176347279. URL <https://projecteuclid.org/journals/annals-of-statistics/volume-17/issue-3/A-Geometric-Interpretation-of-Darroch-and-Ratcliffs-Generalized-Iterative-Scaling/10.1214/aos/1176347279.full>.
- J. N. Darroch and D. Ratcliff. Generalized Iterative Scaling for Log-Linear Models. *The Annals of Mathematical Statistics*, 43(5):1470–1480, 1972. ISSN 0003-4851. URL <https://www.jstor.org/stable/2240069>.
- Robert G. Gallager. *Low-Density Parity-Check Codes*. The MIT Press, September 1963. ISBN 978-0-262-25621-6. doi: 10.7551/mitpress/4347.001.0001. URL <https://direct.mit.edu/books/monograph/3867/Low-Density-Parity-Check-Codes>.
- Alex Goessmann, Janina Schütte, Maximilian Fröhlich, and Martin Eigel. A tensor network formalism for neuro-symbolic AI, January 2026. URL <http://arxiv.org/abs/2601.15442>. arXiv:2601.15442 [cs].
- Martin Grötschel, László Lovász, and Alexander Schrijver. *Geometric Algorithms and Combinatorial Optimization*, volume 2 of *Algorithms and Combinatorics*. Springer, Berlin, Heidelberg, 1993. ISBN 978-3-642-78242-8 978-3-642-78240-4. doi: 10.1007/978-3-642-78240-4. URL <http://link.springer.com/10.1007/978-3-642-78240-4>.
- Wolfgang Hackbusch. *Tensor Spaces and Numerical Tensor Calculus*. Springer Series in Computational Mathematics. Springer-Verlag, Berlin Heidelberg, 2012. ISBN 978-3-642-28026-9. doi: 10.1007/978-3-642-28027-6.
- Michael I. Jordan, editor. *Learning in Graphical Models*. Springer Netherlands, Dordrecht, 1998. ISBN 978-94-010-6104-9 978-94-011-5014-9. doi: 10.1007/978-94-011-5014-9. URL <http://link.springer.com/10.1007/978-94-011-5014-9>.
- Daphne Koller and Nir Friedman. *Probabilistic Graphical Models: Principles and Techniques*. The MIT Press, Cambridge, Mass., 1. edition edition, July 2009. ISBN 978-0-262-01319-2.
- Mattia Miotto and Lorenzo Monacelli. A new pathway to generative artificial intelligence by minimizing the maximum entropy, June 2025. URL <http://arxiv.org/abs/2502.13287>. arXiv:2502.13287 [cs].
- Rahul Rathnakumar, Xuandong Lu, Jiayu Huang, Hao Yan, and Yongming Liu. Bayesian Entropy Neural Networks for physics-aware prediction. *Reliability Engineering & System Safety*, 266:111575, February 2026. ISSN 0951-8320. doi: 10.1016/j.ress.2025.111575. URL <https://www.sciencedirect.com/science/article/pii/S0951832025007756>.
- Stuart Russell and Peter Norvig. *Artificial Intelligence: A Modern Approach, Global Edition: A Modern Approach, Global Edition*. Pearson, Boston, 4 edition, May 2021. ISBN 978-1-292-40113-3.
- C. E. Shannon. A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3):379–423, 1948. ISSN 1538-7305. doi: 10.1002/j.1538-7305.1948.tb01338.x. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/j.1538-7305.1948.tb01338.x>.

- Li Song, Qinghui Guo, Irfan Ali Channa, and Zeyu Wang. A Survey of Maximum Entropy-Based Inverse Reinforcement Learning: Methods and Applications. *Symmetry*, 17(10):1632, October 2025. ISSN 2073-8994. doi: 10.3390/sym17101632. URL <https://www.mdpi.com/2073-8994/17/10/1632>.
- Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, New York, NY, 1st edition edition, September 2018. ISBN 978-1-108-41519-4.
- Martin J. Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge, 2019. ISBN 978-1-108-49802-9. doi: 10.1017/9781108627771.
- Martin J. Wainwright and Michael Irwin Jordan. *Graphical Models, Exponential Families, and Variational Inference*. Now Publishers Inc, 2008. ISBN 978-1-60198-184-4.
- Sangwoong Yoon, Himchan Hwang, Dohyun Kwon, Yung-Kyun Noh, and Frank C. Park. Maximum entropy inverse reinforcement learning of diffusion models with energy-based models. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, volume 37 of *NIPS '24*, pages 24601–24624, Red Hook, NY, USA, December 2024. Curran Associates Inc. ISBN 979-8-3313-1438-5.
- Günter M. Ziegler. Lectures on 0/1-Polytopes. In Gil Kalai and Günter M. Ziegler, editors, *Polytopes — Combinatorics and Computation*, pages 1–41. Birkhäuser, Basel, 2000. ISBN 978-3-0348-8438-9. doi: 10.1007/978-3-0348-8438-9_1. URL https://doi.org/10.1007/978-3-0348-8438-9_1.
- Günter M. Ziegler. *Lectures on Polytopes*. Springer, New York, 1995th edition edition, October 2013. ISBN 978-0-387-94365-7.